Research Project Log Book

IT22187796 | Jayarathna W G U C G | 23 March - 19 October 2026

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Project ID	R26-IT-054
Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing
Student ID	IT22187796
Name	Jayarathna W G U C G
Individual Component	Real-Time Best-Frame Selection and Marker-Free Garment Size Measurement

Summary of the Individual Component

During this month, I worked on understanding my garment size recognition component more clearly. I reviewed its role in the overall garment tracking system and studied how camera-based measurement could reduce manual size-checking work. I identified the main workflow as capturing a garment image, selecting a suitable frame, detecting garment boundaries, measuring dimensions and identifying the size category. I also studied how camera position, garment placement and folding could affect measurement accuracy. This helped me plan a fixed camera setup above the packing table and a simple interface to display garment dimensions and size results.

Date	RP Work
23-03-2026	• Reviewed the project scope and the role of my feature in the overall system. • Identified that my component focuses on detecting garment size using camera-based computer vision methods.
25-03-2026	• Studied how cloth size recognition can support apparel manufacturing. • Focused on identifying garment size automatically instead of manual checking.
27-03-2026	• Analysed the main functions needed for my feature. • Identified garment detection, dimension measurement and size classification as the main functions.

Date	RP Work
30-03-2026	• Studied how garment boundaries can be identified from an image. • Reviewed the importance of width and height measurements for size recognition.

April 2026

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing
Feature	Real-Time Best-Frame Selection and Marker-Free Garment Size Measurement
Period Covered	01 April - 30 April 2026

Date	RP Work
02-04-2026	• Planned the initial workflow of my feature. • Defined the main process as image capture, garment detection, measurement and final size classification.
05-04-2026	• Studied garment appearance in the production environment. • Observed how garment shape, position and placement can affect size recognition.
07-04-2026	• Analysed how camera position can affect garment measurement. • Understood that a fixed camera setup is important for consistent measurements.
08-04-2026	• Studied how garment width and height can be measured from captured images. • Focused on how image-based measurements can represent real garment size.
09-04-2026	• Identified the importance of a proper measurement area. • Planned to capture the garment clearly inside a defined area for size recognition. • Submitted the research paper to ICAC.
11-04-2026	• Worked on the size classification idea. • Reviewed how measured dimensions can be mapped to small, medium, large and extra-large size categories.
12-04-2026	• Refined the feature logic to improve measurement reliability. • Considered errors caused by garment shape changes, unclear boundaries and improper placement.
14-04-2026	• Reviewed real production conditions affecting the feature. • Studied the effect of garment folding, garment position and camera view on recognition.

Date	RP Work
16-04-2026	• Organised the complete workflow of my feature more clearly. • Defined image capture, garment boundary identification, measurement and size classification as the main steps.
18-04-2026	• Reviewed the progress of my feature and the implementation direction. • Built a clearer understanding of the development steps towards component completion.
20-04-2026	• Finalised and submitted the weekly report. • Reviewed my feature progress and the next work direction.
22-04-2026 to 23-04-2026	• Reviewed the frontend prototype for my feature. • Focused on showing garment measurement results clearly in the interface.
25-04-2026	• Reviewed the frontend part of the feature again. • Identified the need to display measured width, height and final size clearly.
27-04-2026 to 28-04-2026	• Started planning the component frontend prototype. • Considered how image capture, measurement and size results should appear step by step.

May 2026

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing
Feature	Real-Time Best-Frame Selection and Marker-Free Garment Size Measurement
Period Covered	01 May - 31 May 2026

Date	**RP Work**
01-05-2026	• Reviewed the feature flow with the frontend prototype idea. • Built a clearer understanding of how my component should appear in the system.
02-05-2026	• Reviewed the overall progress of my component. • Clarified the process of image capture, garment measurement, size classification and result display.
04-05-2026	• Reviewed the current component progress and the next development stage. • Focused on connecting image capture, garment measurement, size classification and result display.
06-05-2026	• Studied how measured garment dimensions can be converted into size categories. • Reviewed mapping width and height measurements to labels such as XS, S, M, L and XL.
08-05-2026	• Analysed how different size standards could be included in the system. • Considered general size categories and UK, US and EUR size outputs. • Reviewed the component objectives for Progress Presentation 1.
10-05-2026	• Reviewed the frontend prototype and measurement result display. • Focused on displaying garment size, dimensions and size-standard results clearly.
11-05-2026	• Prepared the presentation slides. • Practised explaining the research problem, proposed method and current progress.

Date	RP Work
12-05-2026	• Prepared short explanations of image capture, garment detection, measurement, size prediction and result display. • Presented my component at Progress Presentation 1. • Explained the research problem, proposed workflow, frontend prototype and next development steps.
14-05-2026	• Reviewed the dataset requirements for model training and testing. • Identified useful data fields, including garment type, dimensions, folded state and final size label.
15-05-2026	• Reviewed the monthly progress and overall component direction. • Planned camera-based garment detection, dimension measurement, size classification and frontend result display as the development path.
17-05-2026	• Reviewed the research project objectives. • Discussed the next development steps with the supervisor.
20-05-2026	• Reviewed research papers on garment size recognition. • Studied image-based garment measurement methods.
21-05-2026	• Studied different methods for measuring garment dimensions from images.
23-05-2026	• Researched suitable datasets. • Considered methods for collecting garment images.
25-05-2026	• Studied how reference objects or markers can be used to calculate real-world garment measurements.
26-05-2026	• Reviewed garment measurements such as width and length. • Considered how these measurements can support size prediction.
30-05-2026	• Studied how garment measurements can be mapped to S, M, L and XL size categories.

June 2026

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing
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Period Covered	01 June - 30 June 2026

Date	RP Work
02-06-2026	• Reviewed the size-mapping work completed on 30 May and planned the next development stage. • Identified the need for garment videos showing unfolding, incomplete placement and fully visible garments.
06-06-2026	• Prepared the video collection environment and organised the Python, PyTorch and OpenCV development tools. • Checked camera positioning, table coverage and recording quality.
12-06-2026	• Planned how recorded videos would be organised and converted into training images. • Defined suitable frames for garment visibility and measurement.
18-06-2026	• Prepared for garment-site video collection. • Reviewed the need to capture the garment handling process and different unfolding states.
21-06-2026	• Visited the Wallawa garment site to collect video input for the project. • Collected garment footage for subsequent frame selection and measurement development.
23-06-2026	• Visited the Kadawatha garment site to collect video input for the project. • Collected additional footage for dataset preparation and testing.
26-06-2026	• Organised the videos collected so far and checked their suitability for image extraction. • Identified blurred frames, obstructed garments and incomplete garment presentation as cases requiring attention.

Date	RP Work
28-06-2026	• Visited the Mawathagama garment site to collect video input for the project. • Added the collected footage to the project video collection.
29-06-2026	• Visited the Rambukkana garment site to collect video input for the project. • Collected further garment footage for later processing and evaluation.
30-06-2026	• Reviewed the collected garment videos. • Planned frame extraction and the next development steps.

July 2026

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing
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Period Covered	01 July - 31 July 2026

Date	RP Work
02-07-2026	• Organised the video files by collection location. • Checked video quality and garment visibility.
05-07-2026	• Extracted frames from the garment videos. • Removed blurred and unsuitable images.
07-07-2026	• Created a simple frame annotation interface. • Added controls to open videos and review frames.
09-07-2026	• Added READY, NOT_READY and INVALID labels to the annotation interface. • Saved the selected labels.
11-07-2026	• Labelled more frames. • Checked the difference between unfolded garments and incomplete garment placement.
13-07-2026	• Corrected unclear labels and annotation errors. • Combined the annotation records into a CSV file.
15-07-2026	• Prepared the training data for the best-frame classifier. • Checked the input images and labels.
17-07-2026	• Trained the MobileNetV3 baseline model. • Reviewed its predictions for the three frame classes.
20-07-2026	• Checked incorrect READY predictions. • Reviewed difficult frames and corrected dataset issues.

Date	RP Work
22-07-2026	• Fixed model loading and missing-file errors. • Checked the training and prediction file paths.
24-07-2026	• Investigated temporal validation. • Compared predictions from visual and temporal methods.
26-07-2026	• Tested different visual and temporal weights. • Selected the visual-only method for best-frame classification.
28-07-2026	• Reviewed the READY confidence threshold. • Saved the selected baseline model.
30-07-2026	• Built an offline tool to select suitable frames from recorded videos. • Checked the saved images.

August 2026

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing
Feature	Real-Time Best-Frame Selection and Marker-Free Garment Size Measurement
Period Covered	01 August - 31 August 2026

Date	RP Work
01-08-2026	• Created a faster manual frame-selection interface. • Added controls to review and save suitable frames.
03-08-2026	• Processed more recorded videos. • Checked that selected frames showed the full garment.
05-08-2026	• Connected the external webcam. • Checked the camera preview and live video input.
07-08-2026	• Added live frame classification. • Displayed whether the garment frame was ready for capture.
09-08-2026	• Added automatic capture of READY frames. • Checked that unsuitable frames were not saved.
11-08-2026	• Improved capture checks to reduce duplicate images. • Tested different garment positions.
13-08-2026	• Selected images for garment segmentation. • Reviewed visible garment boundaries before annotation.
16-08-2026	• Annotated garment outlines in Roboflow. • Checked the quality of the polygon annotations.
18-08-2026	• Prepared the segmentation dataset. • Corrected export errors and incorrect dataset paths.
21-08-2026	• Visited the Kuliyapitiya garment site. • Collected garment video input for the project.

Date	RP Work
23-08-2026	• Organised the new garment videos. • Selected additional frames for development and testing.
25-08-2026	• Trained a YOLOv8 garment segmentation model. • Checked sample garment masks.
26-08-2026	• Reviewed the collected garment videos, labelled frames and best-frame selection results for the presentation.
27-08-2026	• Reviewed incorrect garment detections. • Identified images that needed better annotation.
28-08-2026	• Prepared the demonstration of the frame-selection tools and live camera workflow. • Reviewed the garment segmentation progress.
30-08-2026	• Updated the presentation slides with the methodology, current results, challenges and remaining work.
31-08-2026	• Tested the demonstration and practised explaining my individual contribution and next development steps.

September 2026

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing
Feature	Real-Time Best-Frame Selection and Marker-Free Garment Size Measurement
Period Covered	01 September - 30 September 2026

Date	RP Work
01-09-2026	• Presented my component at Progress Presentation 2. • Explained video collection, frame labelling, best-frame selection, garment segmentation progress and the remaining measurement and integration work.
03-09-2026	• Reviewed the segmentation dataset. • Added background-only images to improve empty-scene handling.
05-09-2026	• Prepared revised dataset versions. • Checked garment masks and difficult background examples.
07-09-2026	• Refined the overhead camera setup at 100 cm. • Checked the pixel-to-centimetre calibration.
09-09-2026	• Extracted garment masks and contours. • Checked single-garment measurement and empty scenes.
11-09-2026	• Developed garment-specific width and length measurements. • Reviewed T-shirt and shirt measurement positions.
13-09-2026	• Converted pixel distances into centimetres. • Connected the measurements to garment size charts.
15-09-2026	• Created the garment measurement interface. • Added the camera area and garment-type selection.
17-09-2026	• Added the Capture and Measure control. • Connected image capture to the FastAPI measurement service.

Date	RP Work
19-09-2026	• Added measurement lines over the garment image. • Displayed garment width, length and estimated size.
21-09-2026	• Improved feedback when no garment was detected. • Checked invalid and incomplete garment presentations.
23-09-2026	• Added confirmation across three observations. • Kept the result locked until garment removal.
25-09-2026	• Updated the interface to show size counts and total garment counts. • Checked repeated-count prevention.
27-09-2026	• Worked on saved measurements and order-related records using SQLite. • Checked the history display.
28-09-2026	• Integrated the size-measurement interface with the shared desktop application. • Checked communication with the backend.
30-09-2026	• Tested the full capture, measurement and counting flow. • Recorded interface and measurement issues for correction.

October 2026

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing
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Period Covered	01 October - 19 October 2026

Date	RP Work
02-10-2026	• Reviewed the final segmentation dataset and training records. • Organised the evidence for the report.
04-10-2026	• Compared system dimensions with tape measurements. • Recorded width and length differences.
06-10-2026	• Reviewed the measurement error calculations. • Documented the limits of the available test results.
08-10-2026	• Updated the thesis methodology and implementation sections. • Added component diagrams and selected screenshots.
10-10-2026	• Completed the thesis draft, project website and checklist work. • Reviewed the required content and supporting project evidence.
13-10-2026	• Organised the component evidence, including model details, calibration settings and interface screenshots.
16-10-2026	• Reviewed the integrated measurement interface. • Checked the measurement overlay, size result and count display.
18-10-2026	• Organised the development history and logbook content. • Reviewed the remaining thesis and demonstration work.